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EDUCATION

Bachelor of Applied Science in Mechanical Engineering + PEY Co-op

University of Toronto

Sep 2023 - Apr 2028 (expected)

Toronto, ON, Canada

- **Concentrations:** Mechatronics and Bioengineering
- **Intended Minors:** Engineering Business and Robotics
- **GPA:** 3.70/4.00, *Dean's List*
- **Related Coursework:** Mechanical Design, Mechanics of Solids, Engineering Analysis, Computer Fundamentals, App. Fund. of Deep Learning, Corporate Accounting and Finance.

SKILLS & Certifications

Programming: MATLAB, C/C++, JavaScript, Python (PyTorch, TensorFlow)

Tools: Solidworks, ANSYS, Excel, Word, PowerPoint, Anylogic

Professional: Leadership, collaboration, problem-solving, adaptability

Certifications: Certified SolidWorks Associate, Python Specialization - Coursera, Basic Machining - George Brown

PROJECTS

EMG-controlled Robotic Bionic Hand,

Personal Project

Jun 2025 - Jul 2025

Eastern, Saudi Arabia

- Designed a 5-finger **bionic hand** with tendon-like mechanism, prototyped and iterated by 3D printing.
- Collected and trained on **10,000+** EMG samples, testing GRUs, CNNs, and LSTMs on local GPU with **CUDA** acceleration, achieving **83%** accuracy across 4 gesture classes and **200 ms** inference latency.
- Built **computer-vision** teleoperation system using MediaPipe on laptop.
- Implemented surface EMG classification with **TensorFlow Lite** CNN on microcontroller (**C++**).
- Integrated the embedded control on an **ESP32-S3** with PSRAM, driving 5 micro-servos via driver module.

Intro to Embedded Systems,

Independent Project

May 2025 - Jun 2025

Toronto, ON, Canada

- Built circuits and programs for sensors, actuators, and digital interfaces through **15+** Arduino projects.
- Implemented and calibrated analog, I2C, **ultrasonic**, and **infrared sensors** on breadboard circuits.
- Developed a digital thermometer with **$\pm 1^\circ\text{C}$** accuracy using a thermistor and LCD display.
- Programmed an **ESP32** gaming console with **3** games (Snake, Pong, 2048) on a 128×64 OLED display.

Skin Cancer Lesion Classifier, Group Member,

App. Fund. of Deep Learning Course- University of Toronto

Jan 2025 - April 2025

Toronto, ON, Canada

- Trained **CNN** models on **25,000** ISIC 2019 images to classify 8 skin cancer types, achieving **76%** test accuracy.
- Built a **13-million**-parameter CNN as a baseline model to benchmark problem difficulty.
- Integrated **transfer learning** on main model with **PyTorch**'s ResNet-50, improving model performance.
- Applied confusion matrices and Grad-CAM for error analysis and model interpretability.
- Coordinated a 4-member team by managing tasks, deadlines, and meetings to ensure timely project delivery.

Pressure Vessel Testing, Group Member,

Mechanics of Solids Course- University of Toronto

Jan 2025 - April 2025

Toronto, ON, Canada

- Conducted finite element analysis in **ANSYS** Workbench on a cross-sectional pressure vessel model, locating **stress** concentrations under uniaxial loading, analysing heat-maps, and addressing model limitations.
- Conducted **photoelastic** experiments with polarized light to visualize stress patterns and validate simulations.
- Applied **strain gauges** (rosette and uniaxial) to measure stresses and compare with theoretical predictions.
- Carried out **fatigue** testing on steel specimens using a rotating **bending** apparatus to study failure modes.

CNC Router, Team Leader,

Mechanical Engineering Design Course- University of Toronto

Sep 2024 - Dec 2024

Toronto, ON, Canada

- Researched **22** commercial **CNC** models and market gaps to define requirements for beginners and hobbyists.
- Coordinated a 4-member team through concept generation, task allocation, and design iteration.
- Built CAD models in **SolidWorks**, producing a **44**-component assembly optimized for manufacturability.
- Delivered a final design scoring **95%** on rubric for functional and engineering requirements.

Walkable Cities, Team Leader,**Jan 2024 – Apr 2024***Engineering Strategies & Practices II Course- University of Toronto**Toronto, ON, Canada*

- Led a 5-member team through the engineering design process, managing meetings, conflicts, and deadlines.
- Analyzed Galbraith-Bahen crosswalk using traffic, noise, and accessibility data to identify safety gaps.
- Researched municipal, provincial, and **ISO** standards to define design objectives and metrics.
- Simulated traffic-light timing in **AnyLogic**, projecting a **37%** reduction in pedestrian wait times per cycle.
- Created and presented CAD models in **SolidWorks** to instructors and the client for advocacy.

Reversi Board Game, Individual Project,**Feb 2024 – Mar 2024***Computer Fundamentals Course- University of Toronto**Toronto, ON, Canada*

- Developed a **PvP** Reversi game in **C** using 2D arrays and modular functions for game logic and win detection.
- Implemented **PvE** mode with a **minimax** algorithm, **Alpha-Beta** pruning, and a **1-sec** computation limit.
- Competed in a course-wide tournament, ranking **18th** out of 300+ students.

EXTRACURRICULAR EXPERIENCE

NeuroTechUofT**Sep 2025 – Present***Hardware Subsystem - Parkinson's Rehabilitation**Toronto, ON, Canada*

- Incoming member on the NeuroMoveV2 project for parkinson's rehabilitation.

University of Toronto Supermileage Team**Oct 2023 – April 2025***Mechanical Division Member - Urban Concept Car**Toronto, ON, Canada*

- Designed and modified vehicle components in **SolidWorks** through team workshops and tutorials.
- Installed braking system components, including brake-fluid pipes and friction pads, onto the car frame.
- Redesigned the **king-pin** steering mechanism to reduce scrub radius and steering effort, adding self-adjusting geometry for improved handling.
- **Machined** front axle with precision chamfers, reduced diameters, and threads for assembly compatibility.